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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M. Tech (Cyber Security) (CBCS) Degree Examinations

May – June 2025

22CBS231 – CYBER SECURITY ORCHESTRATION, AUTOMATION AND SIMULATION

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and one full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	List and explain the significance of Security Orchestration.	10	L*2	1	1
b)	Why API Security Is Important? Summarize how the APIs concept is used in security orchestration.	10	L2	2	3
2. a)	List and explain, the types of security automation.	10	L2	1	1
b)	Illustrate the security threats related to IP address.	10	L2	1	1
3. a)	Explain seven key steps of VRL filtering process with a neat diagram.	10	L2	1	1
b)	Explain the step by step process to prevent a pass – the – hash attack in Network.	10	L2	1	1
Unit – II					
4. a)	Explain the building blocks of security orchestrion play back.	10	L2	2	3
b)	Demonstrate the SOAR use case of Phishing detection and response with an example.	10	L2	2	3
5. a)	Illustrate the best practice for SOAR Implementation.	10	L2	2	3
b)	Explain the top BAS use cases.	10	L2	3	2
6. a)	Demonstrate the SOAR use case of eendpoint detection and response & detecting suspicious user login from IP address locations.	10	L2	2	3
b)	Explain Evolution of Breach and Attack Simulation (BAS) Devices.	10	L2	3	2
Unit – III					
7. a)	Examine the motivations behind attackers' use of Base 64 algorithms to obfuscate the data.	10	L2	5	2
b)	Explain Detecting Crypto Constants Using FindCrypt2.	5	L2	4	4
c)	Explain briefly Malware Unpacking with diagram.	5	L2	4	4
8. a)	Explain the manual unpacking & Automated Unpacking.	10	L2	5	2
b)	Briefly explain the Memory Acquisition.	5	L2	4	4
c)	Briefly explain Detecting Code Injection with diagram.	5	L2	4	4

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
